

MECG15603/MCCG15603/UECS3153

Statistical Learning

Topic 2b: Supervised Learning:
Logistic Regression & NN

Dr Liew How Hui

Class Arrangement

- Week 9: Lecture 1–3pm (Logistic Regression).
Practical 3–4pm
- Week 10: Lecture 1–3pm. Practical 3–4pm
- Week 11: Lecture 1–3pm. Practical 3–4pm
- Week 12: Oral Presentation in Lecture 1–3pm.
Practical 3–4pm

Recall knowledge from Weeks 1–2: Supervised data has two parts

- input/feature columns $\mathbf{x} = x_1, \dots, x_p$
- output/response column y

Supervised Learning

Problem	Output Y	Arrangement
regression	numerical/ quantitative/ continuous	Week 5–Week 8
classification	categorical/ qualitative/discrete of K classes	Week 9–Week 12

Classification problems with $Y \in \{1, 2, \dots, K\}$ can have a mathematical form

$$Y = (f(\mathbf{X}) + \epsilon \bmod K) + 1.$$

Here, ϵ is a random variable generating integers 1 to K .

Supervised Learning (cont)

A classification problem is called a **binary classification problem** when $K = 2$.

Problem	Prediction $\hat{Y}$	Performance Measurements
regression	$h(X)$, standard deviation	SSE, MSE, RMSE (root mean square error), R^2 , ...
classification	$h(X)$, conditional probability	contingency table/ confusion matrix , accuracy, kappa, ...

Supervised Learning (cont)

Example 1: Let y_i be the actual observed output and $\hat{y}_i$ be the prediction from a predictive model h for the same inputs $\mathbf{x}_i$.

i	$\hat{y}_i$	y_i
1	A	B
2	B	B
3	A	B
4	A	A
5	B	B

Contingency table

		Observed/Actual	
		A	B
Prediction	A	1	2
	B	0	2

In R:

```
Yhat = c("A","B","A","A","B") # first column  
Y     = c("B","B","B","A","B") # second column  
table(Yhat, Y) # Construct contingency table
```

Outline

- 1 Classification Method: Logistic Regression
- 2 Results Interpretation
- 3 Models Comparison
 - Compare Logistic Regression Models with Different Feature Sets
 - Compare to Multinomial Logistic Regression
 - Compare to Artificial Neural Network
- 4 Case Study

Logistic Regression

The *Logistic Regression (LR)* model is a special case of the generalised linear model (GLM) mentioned in Week 7. It is used for **binary classification** and has the form:

$$\ln \frac{\pi}{1 - \pi} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

where $\pi = P(Y = 1 | X_1 = x_1, \cdots, X_p = x_p) = \mathbb{E}[Y]$

(1)

Reference: Wikipedia:Bernoulli Distribution.

The assumption of LR is “the binary data are linearly separable with suitable parameters”. Based on this assumption, a test input $\mathbf{x}$ would get a probability measure.

Logistic Regression (cont)

Rearranging (1) leads to

$$\begin{aligned}\mathbb{P}(Y = 1 | X_1 = x_1, \dots, X_p = x_p) \\&= \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p))} \\&= S(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)\end{aligned}\tag{2}$$

where the logistic/sigmoid function $S(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x}$ has the range $(0, 1)$ for a domain $-\infty < x < \infty$.

Using vector notation, (2) can be expressed compactly as

$$\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x}) = S(\boldsymbol{\beta}^T \tilde{\mathbf{x}})$$

where $\boldsymbol{\beta} = (\beta_0, \dots, \beta_p)$ and $\tilde{\mathbf{x}}_j = (1, \mathbf{x}_j)$.

Logistic Regression (cont)

Given an input $\mathbf{x}$, the LR algorithm provides a prediction as follows based on the conditional probability (assuming the cut-off is 0.5):

$$h(\mathbf{x}) = \begin{cases} 0, & \mathbb{P}(Y = 1|X = \mathbf{x}) < 0.5 \\ 1, & \mathbb{P}(Y = 1|X = \mathbf{x}) \geq 0.5 \end{cases}$$

or based the log-odds (or logit or 'link'):

$$h(\mathbf{x}) = \begin{cases} 0, & \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p < 0 \\ 1, & \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \geq 0 \end{cases}$$

Logistic Regression (cont)

The coefficients β_i are estimated using MLE: Given data $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, we want find the coefficients β_i so that the **likelihood function** of $\beta_0, \dots, \beta_p$ is maximised:

$$\begin{aligned} & L(\beta_0, \dots, \beta_p; y_1, \dots, y_n | \mathbf{x}_1, \dots, \mathbf{x}_n) \\ &= \prod_{i=1}^n \mathbb{P}(Y = y_i | \mathbf{X} = \mathbf{x}_i) \end{aligned} \quad (3)$$

Y is binary and follows a **Bernoulli distribution**.

Logistic Regression (cont)

According to https://en.wikipedia.org/wiki/Bernoulli_distribution,

$Y \sim \text{Bernoulli}(\pi_{\mathbf{x}} = \mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x}))$, then the probability mass function of observing $y \in \{0, 1\}$ is

$$\mathbb{P}(y) = (\pi_{\mathbf{x}})^y (1 - \pi_{\mathbf{x}})^{1-y}.$$

The likelihood for the observation $(\mathbf{x}_i, y_i)$ is

$$\mathbb{P}(Y = y_i | \mathbf{X} = \mathbf{x}_i) = \left(\frac{e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}}{1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}} \right)^{y_i} \left(1 - \frac{e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}}{1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}} \right)^{1-y_i}$$

Logistic Regression (cont)

$$= e^{y_i \tilde{\mathbf{x}}_i^T \boldsymbol{\beta}} \cdot (1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}})^{-y_i} \cdot (1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}})^{-(1-y_i)}$$

where $\boldsymbol{\beta} = (\beta_0, \dots, \beta_p)$ and $\tilde{\mathbf{x}}_i = (1, \mathbf{x}_i)$.

Substituting it into (3), we have

$$\begin{aligned} & L(\beta_0, \dots, \beta_p; y_1, \dots, y_n | \mathbf{x}_1, \dots, \mathbf{x}_n) \\ &= \prod_{i=1}^n (e^{y_i \tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}) (1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}})^{-1}. \end{aligned}$$

Taking natural log leads to log-likelihood:

$$\ln L = \sum_{i=1}^n y_i \tilde{\mathbf{x}}_i^T \boldsymbol{\beta} - \sum_{i=1}^n \ln(1 + e^{\tilde{\mathbf{x}}_i^T \boldsymbol{\beta}}). \quad (4)$$

Theory (cont)

By Calculus Theory,

$$\hat{\beta} = \operatorname{argmax}_{\beta} L = \operatorname{argmax}_{\beta} \ln L \Rightarrow \frac{\partial}{\partial \beta} (\ln L) = \mathbf{0}$$

i.e.

$$\frac{\partial}{\partial \beta} \left(\sum_{i=1}^n y_i \tilde{\mathbf{x}}_i^T \beta - \sum_{i=1}^n \ln(1 + e^{\tilde{\mathbf{x}}_i^T \beta}) \right) = \mathbf{0}.$$

leading to the nonlinear system:

$$\sum_{i=1}^n x_k^{(i)} \left[y_i - \frac{e^{\tilde{\mathbf{x}}_i^T \beta}}{1 + e^{\tilde{\mathbf{x}}_i^T \beta}} \right] = 0, \quad k = 0, 1, \dots, p$$

where $x_0^{(i)}$ is defined to be 1.

Example 1

Consider the following data from ISLR2's Default:

	balance, x_1	default, y
3904	973.9031	No
9146	667.4920	No
5278	1377.4621	No
6930	298.7196	No
7084	919.6724	No
8447	245.3465	No
3024	0.0000	No
8365	1013.2169	Yes
9922	1627.8983	Yes
5210	1711.1691	Yes

Let No=0, Yes=1. The mathematical model for this problem (2) becomes

$$P(Y = 1|X_1 = x_1) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x_1))}$$

Example 1 (cont)

To estimate the coefficients, we need to maximise the likelihood (4):

$$\ln L(\beta_0, \beta_1) = \sum_{i=1}^{10} y_i(\beta_0 + \beta_1 x_{1,i}) - \sum_{i=1}^{10} \ln(1 + e^{\beta_0 + \beta_1 x_{1,i}}).$$

i	$x_{1,i}$	y	y_i	$y_i(\beta_0 + \beta_1 x_{1,i})$	$\ln(1 + e^{\beta_0 + \beta_1 x_{1,i}})$
1	973.9031	No	0	0	$\ln(1 + \exp(\beta_0 + 973.9031\beta_1))$
2	667.4920	No	0	0	$\ln(1 + \exp(\beta_0 + 667.4920\beta_1))$
3	1377.4621	No	0	0	$\ln(1 + \exp(\beta_0 + 1377.4621\beta_1))$
4	298.7196	No	0	0	$\ln(1 + \exp(\beta_0 + 298.7196\beta_1))$
5	919.6724	No	0	0	$\ln(1 + \exp(\beta_0 + 919.6724\beta_1))$
6	245.3465	No	0	0	$\ln(1 + \exp(\beta_0 + 245.3465\beta_1))$
7	0.0000	No	0	0	$\ln(1 + \exp(\beta_0 + 0.0000\beta_1))$
8	1013.2169	Yes	1	$\beta_0 + 1013.2169\beta_1$	$\ln(1 + \exp(\beta_0 + 1013.2169\beta_1))$
9	1627.8983	Yes	1	$\beta_0 + 1627.8983\beta_1$	$\ln(1 + \exp(\beta_0 + 1627.8983\beta_1))$
10	1711.1691	Yes	1	$\beta_0 + 1711.1691\beta_1$	$\ln(1 + \exp(\beta_0 + 1711.1691\beta_1))$
				$\sum_{i=1}^{10} y_i(\beta_0 + \beta_1 x_{1,i})$	$\sum_{i=1}^{10} \ln(1 + e^{\beta_0 + \beta_1 x_{1,i}})$

Example 1 (cont)

We want to find β_0 and β_1 which maximises $\ln L$.
In Calculus, this is the same as minimising $-\ln L$.
Using an initial guess $\beta_0 = 1$ and $\beta_1 = 0$, we are able to find the estimate below using conjugate gradient method:

$$\beta_0 = -5.949928639, \quad \beta_1 = 0.004692262$$

This is very close the estimate obtained using R's `glm()`:

$$\beta_0 = -6.092158330, \quad \beta_1 = 0.004784055$$

The discrepancies are due to numerical formulation.

Theory (cont)

To extend the theory to an input x_i which is categorical, we need to introduce 'dummy variables' to match each levels in the categorical input x_i — **One-Hot Encoding for Categorical Data**.

Example 2: The categorical data (left) to 'dummy variables' (right):

gender	bloodtype
0	A
1	B
1	AB
0	B
0	O

gender1	bloodtypeAB	bloodtypeB	bloodtypeO
0	0	0	0
1	0	1	0
1	1	0	0
0	0	1	0
0	0	0	1

Note: R's `model.matrix`

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Results Interpretation

After we obtain the estimate of the coefficients from the likelihood function:

$$\frac{\partial}{\partial \beta}(\ln L) = \mathbf{0} \Rightarrow \hat{\beta} = \operatorname{argmax}_{\beta} L,$$

The computed results may not fit the data well and we need to answer the following questions.

- 1 Does the model explain the data?
- 2 How does each individual predictor influence the response?

Results Interpretation (cont)

(1) Does the model explain the data?

The statistician's answer, reflected in `glm()`'s output is to compare

- Null deviance = $2(\text{LL}(\mathbf{\text{Saturated Model}}) - \text{LL}(\mathbf{\text{Null Model}}))$ on $\text{df} = \text{df_Sat} - \text{df_Null}$
- Residual deviance = $2(\text{LL}(\mathbf{\text{Saturated Model}}) - \text{LL}(\mathbf{\text{Proposed Model}}))$ on $\text{df} = \text{df_Sat} - \text{df_Proposed}$

The **Saturated Model** is a model that assumes each data point has its own parameters (which means we have n parameters to estimate.)

Results Interpretation (cont)

The **Null Model** assumes the exact “opposite”, in that it assumes one parameter for all of the data points, which means we only estimate 1 parameter.

The **Proposed Model** assumes we can explain the data points with p parameters + an intercept term, so we have $p + 1$ parameters.

If the Null Deviance is really small, it means that the Null Model explains the data pretty well. Likewise for the Residual Deviance. Usually, when null Deviance is much larger than residual deviance, the linear model may explain the data. For prediction purposes, we use the contingency table instead.

Results Interpretation (cont)

(2) How does each individual predictor influence the response?

To answer the question, we analyse the influence of individual predictor to the response using the hypothesis:

$$H_0 : \beta_i = 0 \quad \text{vs} \quad H_1 : \beta_i \neq 0.$$

The Z -statistic of β_i characterises the above hypothesis:

$$Z = \frac{\hat{\beta}_i - 0}{SE(\hat{\beta}_i)}$$

Results Interpretation (cont)

The **square error** in the Z -statistic:

$$SE(\hat{\beta}_i) = [[\mathcal{I}(\beta)]^{-1}]_{(i+1),(i+1)}$$

is the square root of the $(i + 1)$ -th diagonal element of the inverse matrix of the $(p + 1) \times (p + 1)$ **information matrix**:

$$\mathcal{I}(\beta) = \frac{\partial^2}{\partial \beta \partial \beta^T} \left(\sum_{i=1}^n y_i \tilde{\mathbf{x}}_i^T \beta - \sum_{i=1}^n \ln(1 + e^{\tilde{\mathbf{x}}_i^T \beta}) \right) = \sum_{i=1}^n \sigma_i^2 \mathbf{x}_i \mathbf{x}_i^T$$

where $\sigma_i^2 = S(\mathbf{x}_i^T \beta) \cdot (1 - S(\mathbf{x}_i^T \beta))$;

Results Interpretation (cont)

When the number of samples “ n ” is large, the Z-statistic approaches the normal distribution

$$\frac{\hat{\beta}_i - 0}{SE(\hat{\beta}_i)} \sim \text{Normal}(0, 1),$$

according to https://en.wikipedia.org/wiki/Wald_test.

A $(1 - \frac{\alpha}{2}) \times 100\%$ confidence interval for β_i , $i = 1, \dots, p$, can be estimated as

$$\hat{\beta}_i \pm Z_{1-\alpha/2} SE(\hat{\beta}_i).$$

A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter. In this case, $\alpha = 0.05$ and $Z_{1-\alpha/2} \approx 1.96$, therefore, the 95% confidence interval for β_i takes the form

$$[\hat{\beta}_i - 1.96 \cdot SE(\hat{\beta}_i), \hat{\beta}_i + 1.96 \cdot SE(\hat{\beta}_i)]. \quad (5)$$

Results Interpretation (cont)

The interception β_0 is typically not of interest and it only for fitting data to the model.

For β_i where $i = 1, 2, \dots, p$, we have the analysis:

- When Z -statistic is large, p -value is small.
 $\Rightarrow$ null hypothesis should be rejected (when p -value is less than some significance level, e.g. $\alpha=5\%$).
 $\Rightarrow X$ is associated with Y and is a significant predictor.
- When Z -statistic is small, p -value is large.
 $\Rightarrow$ null hypothesis should not be rejected (when (when p -value $> \alpha = 0.05$).
 $\Rightarrow X$ and Y is most likely not related and X is probably an unimportant predictor to Y .

Results Interpretation (cont)

As mentioned in Week 7, a logistic regression model is a special case of GLM where the link function is logit. In R, this is specified using the option 'family=binomial':

```
lr.fit = glm(Y ~ ., data=D, family=binomial)
```

Here `binomial` uses `logit` link (for logistic CDF) by default. Other link options for `binomial` are 'probit', 'cauchit', (corresponding to normal and Cauchy CDFs respectively) 'log' and 'cloglog' (complementary log-log).

Example 3:

```
library(ISLR2)
lr.fit = glm(default ~ balance, data=Default, family=binomial)
print(summary(lr.fit))
```

Results Interpretation (cont)

Example 3: (cont)

Call:

```
glm(formula = default ~ balance, family = binomial, data = Default)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.2697	-0.1465	-0.0589	-0.0221	3.7589

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.065e+01	3.612e-01	-29.49	<2e-16 ***
balance	5.499e-03	2.204e-04	24.95	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2920.6 on 9999 degrees of freedom
Residual deviance: 1596.5 on 9998 degrees of freedom
AIC: 1600.5

Number of Fisher Scoring iterations: 8

Results Interpretation (cont)

Example 3: (cont)

(a) Write down the mathematical formula of the logistic regression model.

Solution

$$\mathbb{P}(Y = 1|X) = \frac{1}{1 + \exp(-(-10.65 + 0.0055 \text{ balance}))}$$

(b) Does the model fit well?

(c) Predict the default probability for an individual with a balance of (i) \$1000, (ii) \$2000.

Exercise.

Results Interpretation (cont)

One reason for the popularity of LR in practice is due to the interpretability of β_i using the notion

https://en.wikipedia.org/wiki/Odds_ratio.

The **odds ratio** (OR) is the ratio between two odds:

$$\text{OR} = \frac{\frac{\mathbb{P}(Y=1|X_i=b)}{\mathbb{P}(Y=0|X_i=b)}}{\frac{\mathbb{P}(Y=1|X_i=a)}{\mathbb{P}(Y=0|X_i=a)}} = \frac{\exp(\cdots + \beta_i \cdot b + \cdots)}{\exp(\cdots + \beta_i \cdot a + \cdots)} = \exp(\beta_i(b - a)).$$

The odds (in the OR) are the ratio of the probabilities of two complementing events:

$$\frac{\mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})}{\mathbb{P}(Y = 0|\mathbf{X} = \mathbf{x})} = \frac{\mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})}{1 - \mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})} = \exp(\tilde{\mathbf{x}}^T \boldsymbol{\beta}). \quad (6)$$

Results Interpretation (cont)

Taking the logarithm of both sides of (6), we obtain (1):

$$\ln \frac{\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x})}{1 - \mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x})} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

The LHS is called the *log-odds* or *logit*, which is linear in X .

For a 1 unit increment in X_i leads to

$$\beta_i > 0 \Rightarrow \text{logit} > 0 \Rightarrow OR > 1 \Rightarrow odds(X_i + 1) > odds(X_i) \Rightarrow \\ \mathbb{P}(Y = 1 | X_i + 1) > \mathbb{P}(Y = 1 | X_i) \text{ (higher prob of } Y \text{ for } X_i + 1)$$

$$\beta_i < 0 \Rightarrow \text{logit} < 0 \Rightarrow OR < 1 \Rightarrow odds(X_i + 1) < odds(X_i) \Rightarrow \\ \mathbb{P}(Y = 1 | X_i + 1) < \mathbb{P}(Y = 1 | X_i) \text{ (lower prob of } Y \text{ for } X_i + 1)$$

Qualitative Predictors

So far the predictors are all assumed numeric. When a predictor (or factor) is **qualitative**, we need to introduce **dummy variable(s)**: For example, the predictor “gender” has two levels 0 (male) and 1 (female), a new variable below is created

$$\text{gender1} = \begin{cases} 1, & \text{if gender} = 1 \\ 0, & \text{if gender} = 0 \end{cases}$$

Therefore, the logistic model is

$$\begin{aligned} & \mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x}) \\ &= \frac{1}{1 + \exp(-(\beta_0 + \cdots + \beta_i \text{gender1} + \cdots))} \end{aligned}$$

Results Interpretation (cont)

The linear algebra theory associated with qualitative predictors are more complex but the result interpretation of the qualitative predictors is also related to the odds ratio, but now, of the the dummy variable(s), for example, “gender1”:

$$\text{OR} = \frac{\frac{\mathbb{P}(Y=1|\text{gender}=1)}{\mathbb{P}(Y=0|\text{gender}=1)}}{\frac{\mathbb{P}(Y=1|\text{gender}=0)}{\mathbb{P}(Y=0|\text{gender}=0)}} = \frac{\exp(\cdots + \beta_i + \cdots)}{\exp(\cdots + 0 + \cdots)} = \exp(\beta_i)$$

Note that 0=male, 1=female, we have

β_i	OR	Relative probability of $\mathbb{P}(Y = 1 \text{gender} = 1)$	Probability to be classified into Class 1
Positive	> 1	Higher	female $>$ male
Negative	< 1	Lower	male $>$ female

Results Interpretation (cont)

Example 4:

Consider the ISLR2's **Default** data. Use R to work on the influence of the student predictor on the output default.

Solution

The R script to fit the logistic model is listed below.

```
library(ISLR2)
lr.fit = glm(default ~ student, data=Default,
             family=binomial)
print(summary(lr.fit))
```

Results Interpretation (cont)

Example 4: (cont)

Call:

```
glm(formula = default ~ student, family = binomial, data = Default)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-0.2970	-0.2970	-0.2434	-0.2434	2.6585

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.50413	0.07071	-49.55	< 2e-16 ***
studentYes	0.40489	0.11502	3.52	0.000431 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2920.6 on 9999 degrees of freedom
Residual deviance: 2908.7 on 9998 degrees of freedom
AIC: 2912.7

Number of Fisher Scoring iterations: 6

Results Interpretation (cont)

Example 4: (cont)

Use the analysis results from R to answer the following questions.

- (a) Find the odds ratio of default for a student with a non-student. Explain.
- (b) Predict the probability of default for (i) student (ii) non-student.

Hint: (i) $\mathbb{P}(Y = 1 | \text{student} = \text{Yes})$; (ii)
 $\mathbb{P}(Y = 1 | \text{student} = \text{No})$

Classroom discussion.

Results Interpretation (cont)

When a qualitative predictor X_i has $K > 2$ levels, $(K - 1)$ **dummy variables** $X_{i.\text{level}2}, \dots, X_{i.\text{level}K}$ are introduced to the logistic regression model

$$\mathbb{P}(Y = 1|\mathbf{X}) = \frac{1}{1 + \exp(-(\beta_0 + \dots + \beta_i^{(2)} x_{i.\text{level}2} + \dots + \beta_i^{(K)} x_{i.\text{level}K} + \dots))}$$

where

$$x_{i.\text{level}k} = \begin{cases} 1, & x_i = \text{level } k, \\ 0, & \text{otherwise,} \end{cases} \quad k = 2, \dots, K.$$

The introduction of $K - 1$ dummy variables is called the “*nearly*” *one-hot encoding*, where the reference variable is implicit. In a **one-hot encoding** all dummy variables are kept.

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Models Comparison

LR Models with two different feature sets are usually considered to be different. They can be compared if the two feature sets come from the same set of data with no missing values.

There are two generalisations of LR models to general (rather than binary) classification problem:

- 1 Multinomial Logit Model
- 2 Feed-forward Neural Network Model

We will also compare a LR model to the generalisations.

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Models Comparison (cont)

As mentioned in Week 7, to compare two trained GLM (LR in particular), we need to look at the deviance (or the closely related AIC, C_p , BIC, etc.).

Since the logistic regression model is a special case of GLM, we can use the deviance to compare models.

This is accomplished using `anova(small.m, large.m)`, where `small.m` is a model with less features, `large.m` is a model with more features. If the deviances' likelihood ratio test (LRT, same as χ^2 test and Rao test) has a p-value less than 0.05, we can conclude that `large.m` is much better at capturing the data than the `small.m`.

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Models Comparison (cont)

A general K -level qualitative response Y cannot be properly handled by the LR model.

https://en.wikipedia.org/wiki/Multinomial_logistic_regression (or Softmax regression) is a generalisation of the LR model:

$$\left\{ \begin{array}{l} \ln \frac{\mathbb{P}(Y = 2 | \mathbf{X} = \mathbf{x})}{\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x})} = \beta_2 \cdot \mathbf{x} \\ \ln \frac{\mathbb{P}(Y = 3 | \mathbf{X} = \mathbf{x})}{\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x})} = \beta_3 \cdot \mathbf{x} \\ \dots\dots\dots \\ \ln \frac{\mathbb{P}(Y = K | \mathbf{X} = \mathbf{x})}{\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x})} = \beta_K \cdot \mathbf{x} \end{array} \right.$$

Models Comparison (cont)

After some algebra, we have the probability of each outcome:

$$\begin{aligned}\mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x}) &= \frac{1}{1 + \sum_{i=2}^K e^{\beta_i \cdot \mathbf{x}}} \\ \mathbb{P}(Y = j|\mathbf{X} = \mathbf{x}) &= \frac{e^{\beta_j \cdot \mathbf{x}}}{1 + \sum_{i=2}^K e^{\beta_i \cdot \mathbf{x}}}, \quad j = 2, \dots, K.\end{aligned}\tag{7}$$

This model requires more data than LR, so when we have little data, this model may not be properly fitted.

Models Comparison (cont)

An implementation of Multinomial LR is available in the `nnet` package:

```
multinom(formula, data, weights, subset, na.action,  
          contrasts = NULL, Hess = FALSE, summ = 0,  
          censored = FALSE, model = FALSE, ...)
```

When $K = 2$, the multinomial LR is just the usually logistic regression model and we will explore this in the practical.

Models Comparison (cont)

In Python, the “Logistic Regression” is actually a generalisation to the **elastic net** instead of the LR we discussed:

```
class sklearn.linear_model.LogisticRegression(penalty='l2', *,
        dual=False, tol=0.0001, C=1.0, fit_intercept=True,
        intercept_scaling=1, class_weight=None, random_state=None,
        solver='lbfgs', max_iter=100, multi_class='auto', verbose=0,
        warm_start=False, n_jobs=None, l1_ratio=None)
```

When $C = \infty$, it approaches the LR. The LR and multinomial LR are properly implemented in Python as Logit and MNLogit in `statsmodels.discrete.discrete_model`.

Outline

- 1 Classification Method: Logistic Regression
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Models Comparison (cont)

Feed-forward Artificial Neural Networks (ANN) or multi-layer perceptrons (MLP), “include” LR and multinomial LR as special cases.

A multi-layer feed-forward ANN with input $\mathbf{x}_i \in \mathbb{R}^p$ and output is $\mathbf{y}_i \in \mathbb{R}^m$:

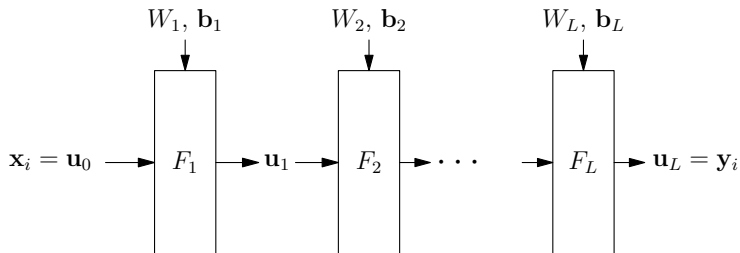
$$\begin{aligned}\mathbf{u}_1 &= F_1(W_1\mathbf{u}_0 + \mathbf{b}_1), & \mathbf{u}_0 &= \mathbf{x}_i \\ \mathbf{u}_2 &= F_2(W_2\mathbf{u}_1 + \mathbf{b}_2) \\ &\dots\end{aligned}\tag{8}$$

$$\hat{\mathbf{y}}_i = \mathbf{u}_L = F_L(W_L\mathbf{u}_{L-1} + \mathbf{b}_L).$$

where L is the number of layers of ANN (with $L - 1$ hidden layers).

Models Comparison (cont)

Horizontal pictorial representation:



ANN is a class of models. In practice, we need to decide on how many layers to choose and for each layer, how many nodes to choose. Typical configurations are rectangular (each layer has same number of nodes) or sandglass shape (wide to narrow to wide).

Models Comparison (cont)

The output $\hat{\mathbf{y}}$ can be a sequence of probabilities $P(Y = j|\mathbf{x})$. It can also be a sequence numeric values (e.g. when it is applied in regression problems).

The algorithm to estimate the parameters W_ℓ and $\mathbf{b}_\ell$ for the layer $\ell = 1, \dots, L$ is the improvement of back-propagation algorithm:

- 1 $t = 0$;
- 2 Using the guess parameters $W_\ell^{(t)}$, $\mathbf{b}_\ell^{(t)}$, calculate all the intermediate states

$$\mathbf{u}_\ell^{(t)} = F_\ell(W_\ell^{(t)} \mathbf{u}_{\ell-1}^{(t)} + \mathbf{b}_\ell^{(t)})$$

and the output $\hat{\mathbf{y}}_i$;

Models Comparison (cont)

- 3 The output layer

$$\delta_L = \hat{\mathbf{y}}_i - \mathbf{y}_i$$

- 4 Back-Propagation (roughly): For ℓ from L to 1, do

$$\delta_{\ell-1} = \frac{\partial F_\ell}{\partial \mathbf{W}_\ell}(\mathbf{u}_{\ell-1}^{(t)}) \delta_\ell$$
$$\mathbf{W}_\ell^{(t+1)} = \mathbf{W}_\ell^{(t)} + \alpha \times \mathbf{u}_{\ell-1}^{(t)} \times \delta_{\ell-1}$$

- 5 $t = t + 1$ and go to step 2.

Models Comparison (cont)

When $L = 1$, we obtain a

<https://en.wikipedia.org/wiki/Perceptron>:

$$\mathbf{y} = \mathbf{u}_1 = F_1(W_1 \mathbf{x}_i + \mathbf{b}_1). \quad (9)$$

We can see that when $m = 1$, $F_1(x) = S(x)$, we obtain the LR. When $m = K - 1$ ($K \geq 2$), we obtain the multinomial LR (which is how `nnet::multinom` was implemented).

Models Comparison (cont)

When $L = 2$, we obtain an ANN with a single hidden-layer.

$$\begin{aligned}\mathbf{u}_1 &= F_1(W_1\mathbf{x}_i + \mathbf{b}_1) \\ \mathbf{y} = \mathbf{u}_2 &= F_1(W_2\mathbf{u}_1 + \mathbf{b}_2).\end{aligned}\tag{10}$$

This is implemented in R's `nnet` package as

```
nnet(x, y, weights, size, Wts, mask,  
     linout = FALSE, entropy = FALSE, softmax = FALSE,  
     censored = FALSE, skip = FALSE, rang = 0.7, decay = 0,  
     maxit = 100, Hess = FALSE, trace = TRUE, MaxNWts = 1000,  
     abstol = 1.0e-4, reltol = 1.0e-8, ...)
```

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Case Study 1: Simple Model Comparison

Example 5: Given the info of a fitted model below.

```
Call: glm(formula=default~balance+income+student, family=binomial,
          data=Default)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.4691	-0.1418	-0.0557	-0.0203	3.7383

Coefficients:	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.087e+01	4.923e-01	-22.080	< 2e-16 ***
balance	5.737e-03	2.319e-04	24.738	< 2e-16 ***
income	3.033e-06	8.203e-06	0.370	0.71152
studentYes	-6.468e-01	2.363e-01	-2.738	0.00619 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2920.6 on 9999 degrees of freedom

Residual deviance: 1571.5 on 9996 degrees of freedom

AIC: 1579.5

Number of Fisher Scoring iterations: 8

Case Study 1 (cont)

Discuss the results involving the coefficients, odds and significance of each variable.

Solution

Coefficients: $\beta_0 = -10.8690$, $\beta_1 = 0.0057$,
 $\beta_2 = 3.033 \times 10^{-6}$, $\beta_3 = -0.6468$.

Significance: Based on the p -value, we find that balance and student are significant while income is probably insignificant (according to the default $\alpha = 0.05$).

Odds: The odds of the default increases with the balance and income but students has a lower odds compare to non-students.

Case Study 1 (cont)

Performing an ANOVA from NULL model to full model using χ^2 -test, we obtain

Analysis of Deviance Table

Model 1: default ~ 1

Model 2: default ~ student

Model 3: default ~ student + balance

Model 4: default ~ student + balance + income

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)	
1	9999	2920.7				
2	9998	2908.7	1	11.97	0.0005416	***
3	9997	1571.7	1	1337.00	< 2.2e-16	***
4	9996	1571.5	1	0.14	0.7115139	

“Model 3” is the best model.

Case Study 2

Example 6: Given the following results from the analysis of credit card applications approval dataset using logistic regression model.

```
glm(formula=Approved~., family=binomial, data=d.f.train)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.6796	-0.5477	0.2681	0.3316	2.4501

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.1379649	0.5744168	5.463	4.68e-08	***
Maleb	-0.1758676	0.3229541	-0.545	0.5861	
Age	0.0001318	0.0142338	0.009	0.9926	
Debt	0.0042129	0.0298740	0.141	0.8879	
YearsEmployed	-0.1023132	0.0582368	-1.757	0.0789	.
PriorDefaultt	-3.6614227	0.3659226	-10.006	< 2e-16	***
Employedt	-0.2500687	0.4013495	-0.623	0.5332	
CreditScore	-0.1098142	0.0644360	-1.704	0.0883	.
ZipCode	0.0011958	0.0009540	1.253	0.2100	
Income	-0.0004544	0.0001966	-2.311	0.0209	*

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 625.90 on 454 degrees of freedom
Residual deviance: 294.33 on 445 degrees of freedom
(27 observations deleted due to missingness)
AIC: 314.33

Case Study 2 (cont)

Example 6: (cont)

where the output `Approved` is either positive (represented as 0) and negative (represented as 1) and the features

- `Male` is categorical with `a=Female`, `b=Male`;
- `PriorDefault` is categorical with `f=false`, `t=true`;
- `Employed` is categorical with `f=false`, `t=true`;
- `Age`, `Debt`, `YearsEmployed`, `CreditScore`, `ZipCode`, `Income` are continuous variables.

Case Study 2 (cont)

(i) Write down the mathematical expression of the logistic model for the given data with the coefficient values rounded to 4 decimal places.

(i) Solution

The logistic model is

$$\mathbb{P}(\text{Approved} = 1 | \mathbf{X}) = \frac{1}{1 + e^{-(3.1380 + \mathbf{w}^T \mathbf{X})}}$$

$$\begin{aligned} \mathbf{w}^T \mathbf{X} = & -0.1759 \text{Male} + 0.0001 \text{Age} + 0.0042 \text{Debt} - 0.1023 \text{YearsEmployed} \\ & - 3.6614 \text{PriorDefault} - 0.2501 \text{Employed} - 0.1098 \text{CreditScore} \\ & + 0.0012 \text{ZipCode} - 0.0005 \text{Income} \end{aligned}$$

Case Study 2 (cont)

(ii) By calculating the probability of the credit card application being approved for a male of age 22.08 with a debt of 0.83 unit who has been employed for 2.165 years with no prior default and is currently unemployed, has a credit score 0 and a zip code 128 with income 0, find the **probability** of credit card applications approval and determine if the approval is positive or negative (using the cut-off of 0.5).

Case Study 2 (cont)

(ii) Solution

First, we calculate

$$\begin{aligned}\mathbf{w}^T \mathbf{X} &= -0.1759(1) + 0.0001(22.08) + 0.0042(0.83) - 0.1023(2.165) \\ &\quad - 3.6614(0) - 0.2501(0) - 0.1098(0) \\ &\quad + 0.0012(128) - 0.0005(0) \\ &= -0.2380855\end{aligned}$$

The probability of the credit card application being 'negatively' approved,

$$\mathbb{P}(\text{Approved} = 1 | \mathbf{X}) = \frac{1}{1 + \exp(-(3.1380 - 0.2380855))} = 0.9478$$

Since the probability is more than 0.5, the approval is **negative**.

Case Study 2 (cont)

(ii) Advice for Final Exam

If you are not able to write the formula $\mathbf{w}^T \mathbf{X}$ in the previous correctly, try to for a table based on the given information:

Feature, X_i	Value, x_i	$\beta_i x_i$ (one-hot)
Male	male=b	-0.1758676×1
Age	22.08	0.0001318×22.08
Debt	0.83	0.0042129×0.83
YearsEmployed	2.165	-0.1023132×2.165
PriorDefault	no=f	$0 = -3.6614227 \times 0$
Employed	unemployed=f	$0 = -0.2500687 \times 0$
CreditScore	0	0
ZipCode	128	0.0011958×128
Income	0	0
	Sum:	-0.237906427
	$\beta_0 + \mathbf{w}^T \mathbf{X}$	$3.1379649 - 0.237906427 = 2.900058473$

Case Study 2 (cont)

(iii) Calculate the odds ratio for the approval being negative with the prior default to be true against the prior default to be false. Infer the likelihood of getting a negative approval based on the prior default.

(iii) Solution

The odds ratio for the approval with respect to prior default is

$$\frac{\frac{\mathbb{P}(\text{Approved}=1|\text{PriorDefault}=t)}{1-\mathbb{P}(\text{Approved}=1|\text{PriorDefault}=t)}}{\frac{\mathbb{P}(\text{Approved}=1|\text{PriorDefault}=f)}{1-\mathbb{P}(\text{Approved}=1|\text{PriorDefault}=f)}} = \frac{\exp(-3.6614227 \times 1)}{\exp(-3.6614227 \times 0)} = 0.02569593$$

Someone with a prior default has a lower likelihood to get a negative approval compare to someone without a prior default.

Case Study 3

Example 7:

(a) The human resource department would like to determine potential employees for promotion. You have collected some data from previous employee promoting records as described below:

exp	Number of years of experience working in the company
sal_mth	Average monthly salary in last 12 months
sal_yr	Yearly salary in last 12 months
pjt	Is there any project involved? [Yes; No]
dpmt	Department [A; B; C; D]
emp_id	Employee ID
promote	Is the employee getting promoted? [Yes=1; No=0]

Case Study 3 (cont)

A logistic regression has been constructed to predict the promotion of an employee. Table Q2(a) shows parts of the results of the logistic regression.

	Coefficient	<i>P</i> -value
Intercept	0.0035	$< 2e-16$
exp_yr	0.7124	$< 2e-16$
sal_mth	-0.0212	0.0057
sal_yr	-0.0363	0.0086
pjt_Yes	0.0330	0.2479
dpmt_B	1.0447	0.0002
dpmt_C	-1.5318	6.87e-05
dpmt_D	2.1539	0.0017
emp_id	-0.0279	0.5245

Table Q2(a)

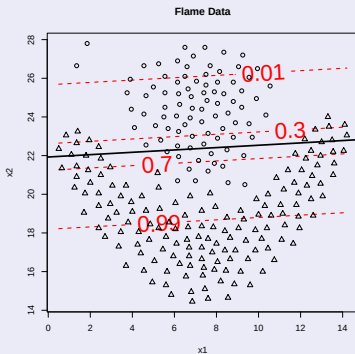
Case Study 3 (cont)

- ❶ Write the logistic regression model that compute the probability that an employee get promoted, $\mathbb{P}(Y = 1)$.
- ❷ Calculate the odds and compare the probability of promotion for employee with 7 years of working experience and an employee with 2 years of working experience.
- ❸ Calculate the odds and compare the probability of promotion for employee in different departments. Arrange the probability of promotion of department from lowest to highest.

Case Study 4

ROC Example

For the “flame” data, the “boundary” of the classifier is shown in the left figure below as the solid line:



Case Study 4 (cont)

ROC Example continue

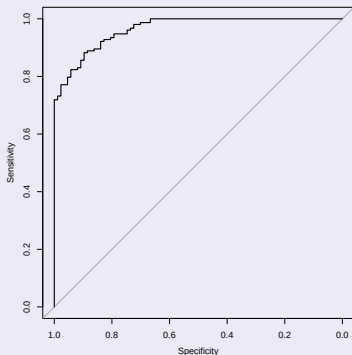
The dashed lines correspond to different “cut-off” 0.01, 0.3, 0.7 and 0.99.

The ROC curve can be understood as the result of varying the “cut-off” and calculating the “sensitivity” (TPR) and “specificity” mentioned in Topic 1. If we calculate out, we have

Predicted	0.01		0.3		0.7		0.99	
	1	2	1	2	1	2	1	2
1	19	0	64	6	79	23	87	80
2	68	153	23	147	8	130	0	73
	TPR = 0.2184	FPR = 0	0.7356	0.0392	0.9080	0.1503	1	0.5229

Case Study 4 (cont)

ROC Example continue



Case Study 5

Large Language Models (LLM) are complex neural networks with token encoding, attention, feed-forward NN known as transformer architecture.

- Sentences are encoded into “tokens + positions”
- Attentions involves multiplication of matrices to get the probabilities of “relations” between tokens.
- Output : Probability of tokens, only the highest probability will be selected,

$$\operatorname{argmax}_i P(Y = \text{token}_i | X_1, \dots, X_n = \text{earlier tokens})$$

where Y is the next token.

Case Study 5 (cont)

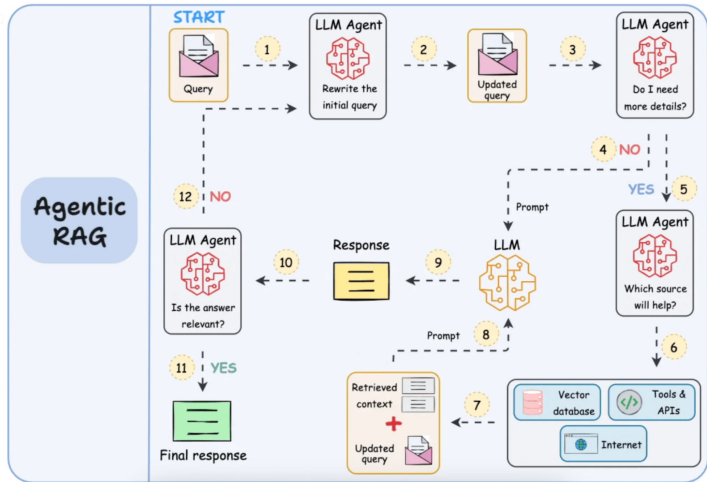
Application of LLMs:

- 1 Agent uses Skills or Sub-agents or LLM or MCP (Model Context Protocol) to generate a sequence of instructions to get and organise data. E.g. OpenClaw
- 2 Skills uses MCP to get data and perform actual work.

Note: For software engineers, agents are mostly NodeJS-based software that communicates with Browser, LLM, sub-agents and uses skill configurations and protocols like MCP to get things done.

Case Study 5 (cont)

Typical Agentic Retrieval-Augmented Generation (RAG)
(Source: <https://atlan.com/know/what-is-rag/>)



On Assignment 2

Form a three-member to five-member group and work on Assignment 2 in Week 9 & Week 10. Inform lecturer the group members & Data of Interest.

Start reading the assignment and exploring the data in the assignment based on what you have learned from the Week 1 to Week 9 practicals.

Oral Presentation for Assignment 2 is in Week 12. First group to register presents last; ...; Last group to register presents first.

By Week 10, the lecturer will randomly generate groups of 3 to 5 members for those who have not formed any group.

Lab Practice on Classification Method

`prac_cls1-lr.R`